A FINITE TYPE MAP FROM AN AFFINE SCHEME TO A QUASI-SEPARATED ALGEBRAIC SPACE THAT ADMITS NO IMMERSION INTO AFFINE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The final 'which is not an immersion' is the author's own conclusion drawn from the displayed two-branch scheme, kept verbatim; the section's positive scheme statement (2402-2406) is provenance and is not part of the delivery.

Editorial note. The label lemma-cannot-embed-into-affine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2424-2436: Y = A^1_k mapping to X = A^1_k/R where R = {(t, t)} union {(t, -t), t != 0}; an immersion Y -> A^n_X over X would be given by regular functions f_1, ..., f_n on Y; the fibre product of Y with the covering A^{n+1}_k -> A^n_X is the scheme {(f_1(t), ..., f_n(t), t)} union {(f_1(t), ..., f_n(t), -t), t != 0} with the obvious morphism to A^{n+1}_k, which is not an immersion
- The context closing sentence: this gives a counter example with X quasi-separated
- The positive backdrop (section opening, 2402-2406): for schemes, a locally finite type morphism from an affine scheme immerses into some A^n_X (Morphisms, quasi-affine-finite-type-over-S) - provenance only

A second counter example is $Y = \mathbf{A}^1_k \to X = \mathbf{A}^1_k/R$ with $R = \{(t,t)\} \amalg \{(t,-t), t \neq 0\}$. Namely, in this case the morphism $Y \to \mathbf{A}^n_X$ would be given by some regular functions $f_1, \ldots, f_n$ on Y and hence the fibre product of Y with the covering $\mathbf{A}^{n+1}_k \to \mathbf{A}^n_X$ would be the scheme

$$\{(f_1(t), \ldots, f_n(t), t)\} \amalg \{(f_1(t), \ldots, f_n(t), -t), t \neq 0\}$$

with obvious morphism to $\mathbf{A}^{n+1}_k$ which is not an immersion. Note that this gives a counter example with X quasi-separated.

Lemma 0.1. *There exists a finite type morphism of algebraic spaces $Y \to X$ with Y affine and X quasi-separated, such that there does not exist an immersion $Y \to \mathbf{A}^n_X$ over X.*

Proof. See discussion above. □

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.